

Halemane's Theory on the Collatz ($3x+1$) System

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Abstract

This paper presents the Halemane's Theory on the Collatz ($3x+1$) System; that is, the Collatz-Thwaites-Ulam-Hasse-Syracuse-Kakutani (CTUHSK) System; which asserts the convergence of the Collatz Sequence to the trivial-cycle $\{(1 \Leftarrow 2 \Leftarrow 4)\}$; thereby proving the Collatz Conjecture, a long-standing unresolved problem.

The transitive closure of the subset $\{1,5,3\}$ under the inverse Collatz function is shown to be the set of all natural numbers, asserting that the Collatz sequence starting from any given natural number converges to the trivial-cycle.

An exact representation of the inverse Collatz function is achieved by a bounded finite small-sized ideal-based graded-algebraic-filtration-structure defined on a modulo-2 quotient-semiring generated by the pair of roots -1 and 3.

The trivial-cycle of the Collatz-map is bypassed through an initialization-phase for this graded-algebraic-filtration-structure, starting directly with a 5-layered structure with the modulo-16 coprime-layer as its topmost layer. This also facilitates a clear distinction among the six distinct possible combinations of the modulo-4 residue-classes and the modulo-6 residue-classes that are associated with any positive odd number; which is blurred in smaller structures.

A layer-shifting global-affine-transformation, defined by the inverse Collatz function; results in a Euclidean shift to the topmost coprime-layer of this filtration structure; avoiding the modulo-multiple-layer (zero-layer) and all the intermediate nilpotent-layers with nilpotent-elements (dead-end zero-divisors).

We establish that the transitive closure of the subset $\{1,5,3\}$ under the inverse Collatz function is indeed the entire set of all natural numbers; covering all the relevant (modulo-3 & modulo-6) modular-residue-classes and also covering all the possible valid triplet-combinations of (1) input-values (2) operations and (3) output-values; governed by the modular-periodicity characteristics (resulting in the self-similar symmetry structure) of the Collatz system.

Halemane-Conjecture states that the maximum number of odd $(3x+1)$ operations required to reach the trivial-cycle $\{(1 \Leftarrow 2 \Leftarrow 4)\}$; starting from any given positive integer greater than one and moving along the Collatz sequence; is limited by that given number itself; with the triad $\{(31 \Leftarrow 41 \Leftarrow 27)\}$ as an exceptional limiting case.

Keywords: Binary-Exponential-Ladder; Modular-Periodicity;
 Collatz-Thwaites-Ulam-Hasse-Syracuse-Kakutani (CTUHSK) System;
 Halemane's Theory of the CTUHSK System; Halemane-Conjecture.

AMS MSC Mathematics Subject Classification: 11B83; 11Y55; 93A05; 93A99.

1. Introduction

The Collatz-Thwaites-Ulam-Hasse-Syracuse-Kakutani (CTUHSK) Conjecture, simply referred to as the *Collatz Conjecture* [§1] states that the Collatz Sequence, also referred to as the Collatz $(3x+1)$ Sequence, converges to the trivial-cycle, that is, $\{(1 \Leftarrow 2 \Leftarrow 4)\}$, starting from any positive integer.

Lagarias [§3]&[§4] gives an exhaustive annotated bibliography, whereas Lagarias [§5]&[§6] gives an elaborate overview of the Collatz Problem, also referred to as the “ $3x+1$ problem”, referring to it as “The Ultimate Challenge”. Guy [§2] has been somewhat pessimistic in advising researchers “Don't Try to Solve These Problems”. Lagarias [§5]&[§6] refers to the statement by the most revered Mathematician and Number Theory Expert Paul Erdos, who said that - "Mathematics is not yet ready for such problems" - about the Collatz Conjecture. The Collatz $(3x+1)$ Sequence is indeed a highly complex chaotic system defined or generated with one of the

simplest deterministic well-defined equations, that too, just two in number. We will not get into any more of the reported literature except the problem definition; but will simply focus on a rather shockingly simple approach to solve this amazing long-standing unresolved problem.

This paper presents an algebraic model for the Collatz $(3x+1)$ system; asserting the convergence of the Collatz sequence to the trivial-cycle $\{(1 \Leftarrow 2 \Leftarrow 4)\}$.

2. Algebraic Structure for the Inverse Collatz Function

The inverse of the condensed compact Collatz function, defined on the set of positive odd numbers, is given by:

$$\begin{aligned} \text{if } (n \equiv 1 \pmod{6}) \text{ then } & P(n) := g(n) := (n \cdot 2^w - 1)/3; \\ \text{else if } (n \equiv 5 \pmod{6}) \text{ then } & P(n) := h(n) := (n \cdot 2^v - 1)/3; \end{aligned} \quad [\text{Eqn.2}]$$

with w being a positive even exponent and v being a positive odd exponent; where P refers to the Collatz predecessor operator.

Note that the Collatz predecessor is a one-to-many mapping.

The algebraic structure for the inverse Collatz function has the following characteristics.

2.1 BASE SEMIRING AND IDEAL

Start with the semiring of non-negative integers, $R = (\mathbb{Z}_{\geq 0}, +, \cdot)$;

The modulo-2 quotient semiring is constructed by considering the ideal $I = \langle 2 \rangle$ defined by the modulo-base 2.

2.2 PRIMITIVE-ROOT

The element 3 serves as a primitive-root modulo-3, because $3 \equiv 1 \pmod{2}$;

allowing us to achieve the division by 3 operation within the semiring structure.

Every division by 3 (whenever applicable, as per the inverse Collatz function) is achieved through multiplication by the modular inverse of the primitive-root in the specific quotient-semiring; for example, multiplying by 11 in modulo-16 semiring.

2.3 DIVISIBILITY-CLASSES

The divisibility-classes based on the modulo-2 base (used for the creation of the quotient semiring) defines the subring layers in the filtration scheme.

2.4 IDEAL-BASED FILTRATION SCHEME

To define a hierarchical filtration scheme, we create a tower of subrings or ideals, corresponding to the divisibility-order with respect to the modulo-base 3; the bottommost being the zero-layer or the all-absorbing-layer. Each higher layer is

created to accommodate for the possible valid values of the binary exponents w and v in [Eqn.2] above as required in the definition of the inverse Collatz function; the topmost layer being the units or the coprime-layer forming the principal-orbit. The quotient semirings defined with respect to modulus base 2^k characterize layer k with divisibility-order k in this filtration scheme. The primitive-root 3 generates the multiplicative structure within any given subring; and the elements transition down the filtration hierarchy by exact division by 2; although the inverse Collatz function does not have any operation of division by 2.

Note that the process of repeated division by the primitive-root 3, within the subring of any given filtration layer, produces two partitions of the subring consisting of alternating elements of the orbit in the subring; one belonging to the modulo-4 residue class $[1]_4$ and the other belonging to the modulo-4 residue class $[3]_4$; both being of exactly equal size.

2.5 EUCLIDEAN EXPANSION SHIFT TO THE TOPMOST COPRIME LAYER

The inverse Collatz operators $g(n)$ & $h(n)$ as in [Eqn.2] above; act as the filtration shifting global affine transformations, achieving a Euclidean expansion shift to a higher layer; while also avoiding the modulo multiple layer (zero-layer) and all the intermediate nilpotent-layers with nilpotent elements (dead-end zero-divisors) like $(4m-2)$ and $(4m-0)$. This upward jump may be either (1) an even number w of binary exponential hops (based on the chosen value of w) in case of $g(n)$; or (2) an even number w of binary exponential hops (based on the chosen value of w) in case of $h(n)$. The valid values for w and v are determined by the modulo-6 residue class of the operand element, as specified in [Eqn.2] above. When either of these inverse

Collatz odd operators of [Eqn.2] act on a valid operand element in the current topmost coprime-layer, or even when $g(n)$ of [Eqn.2] acts on a valid operand element in a penultimate (just beneath the topmost) layer; we create a new topmost coprime-layer with a higher modulus base of 2^{k+1} instead of the earlier 2^k ; leading to a dynamically evolving expansion of the filtration scheme.

2.6 INITIALIZATION PHASE TO BYPASS THE TRIVIAL CYCLE

The trivial-cycle $\{(1 \Leftarrow 2 \Leftarrow 4)\}$ can be bypassed by an initialization-phase for this filtration scheme, starting directly with the modul-16 coprime-layer.

2.7 AN ELEGANT PROOF OF THE COLLATZ CONJECTURE

The transitive-closure of the inverse Collatz System is exactly represented by the transitive-closure of this dynamically evolving algebraic system which is indeed the complete set of natural numbers. This design for the system-dynamics-framework by itself provides a definitive-proof of the Collatz $(3x+1)$ Conjecture, establishing that the entire inverse Collatz Map is exactly represented by this dynamically-evolving graded-algebraic-structure defined on the set of all natural numbers; neither missing any natural number nor including any extraneous-elements; providing the necessary & sufficient condition.

3. Recommended Reading

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4. Declaration Regarding Affiliation and Funding

I, Dr(Prof) Keshava Prasad Halemane, hereby declare that I am a Professor retired as on 2017JAN31 from National Institute of Technology Karnataka Surathkal India, and I am not affiliated to any institution or organization or corporation or any other agency or whatever. This research work has been conducted entirely by me on my own as an Independent Researcher, and that I have not received any funding from any source other than my own savings, and I do not have any obligations or encumbrances of any kind, neither financial nor legal nor of any other kind, regarding the contents of the manuscript - of which I am the original author and creator. Also, I hereby declare there is no conflict of interests.

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6. Dedication

To my ಅಜ್ಜ (ajja) Karinja Halemane Keshava Bhat & ಅಜ್ಜಿ (ajji) Thirumaleshwari, ಅಪ್ಪ (appa) Shama Bhat & ಅಮ್ಮ (amma) Thirumaleshwari, for their *teachings through love, that quality matters more than quantity*; to my wife Vijayalakshmi for her *ever consistent love & support*; to my daughter [Sriwidya.Bharati](#) and my twin sons [Sriwidya.Ramana](#) & [Sriwidya.Prawina](#) for their *love & affection*.